

Archan Das

Full-Stack ML Research Engineer

San Francisco, CA

📞 224-355-8973

✉ creeksidedas@gmail.com

Education

- May 2025** **M.S. in Computer Science (Research)**, *Carnegie Mellon University, School of Computer Science*, Pittsburgh, PA
Relevant Courses: Deep Learning Systems (10-714), Data Mining on Multimedia Databases (15-826), LLM Systems (11-868), Multimodal ML (11-777), Advanced NLP (11-711)
- May 2024** **B.S. in Computer Science**, *Carnegie Mellon University, School of Computer Science*, Pittsburgh, PA, GPA: 4.0
Relevant Courses: Introduction to Machine Learning (10-701), Compilers (15-411), Introduction to Deep Learning (11-485)

Experience

- Aug 2025 – present** **Senior AI Research Engineer**, *Archer Aviation*
- Full stack ML engineer building AI products in the aviation space
 - Training + inference on audio-language models for aviation domain
 - Built backend (vLLM, Rust, NATS) and on-device (iOS, CoreML, MLX) inference platforms for ASR and LLM models, scaled to handle every airport in the US
 - Technical lead for MLOps platform designed to accelerate research and product handoff for multiple production models
- Sep 2023 – Dec 2023** **Software Engineer Intern**, *Profound*
- Helping build the next generation of brand analytics for LLMs.
 - Collecting and managing large datasets.
- May 2023 – Aug 2023** **Software Engineer Intern**, *Artera*
- Performed full-stack development at a health tech startup.
 - Built backend data infrastructure using Terraform and AWS.

Research and Projects

- 2026** **Personal PM** Using on-device multimodal LLMs to track my activity throughout the day and nudge towards better productivity
- 2024 – 2025** **Analyzing Debugging Behavior Using Programming Process Data** MS Thesis. Using IDE log data collected from an introductory CS course to reconstruct phases of the mental model of the debugging process.
- 2024** **Bus Route Optimization** Designed and simulated Pittsburgh's bus network using public data; trained a deep reinforcement learning algorithm to dynamically optimize route planning based on rider demand.
- 2024** **Gas Price Forecasting** Utilized economic time series data from the BEA to generate projections for gas prices and inflation metrics in Pittsburgh, employing Bayesian VAR state space methods in R and MATLAB.

Skills

- Programming** Rust, Python, PyTorch, Swift, CUDA C++
- Tools** Docker, GCP, AWS, Kubernetes, Terraform, NATS